Synopsis

Project Title: Student Academic Performance Prediction Using Machine Learning.

Problem Id: 27

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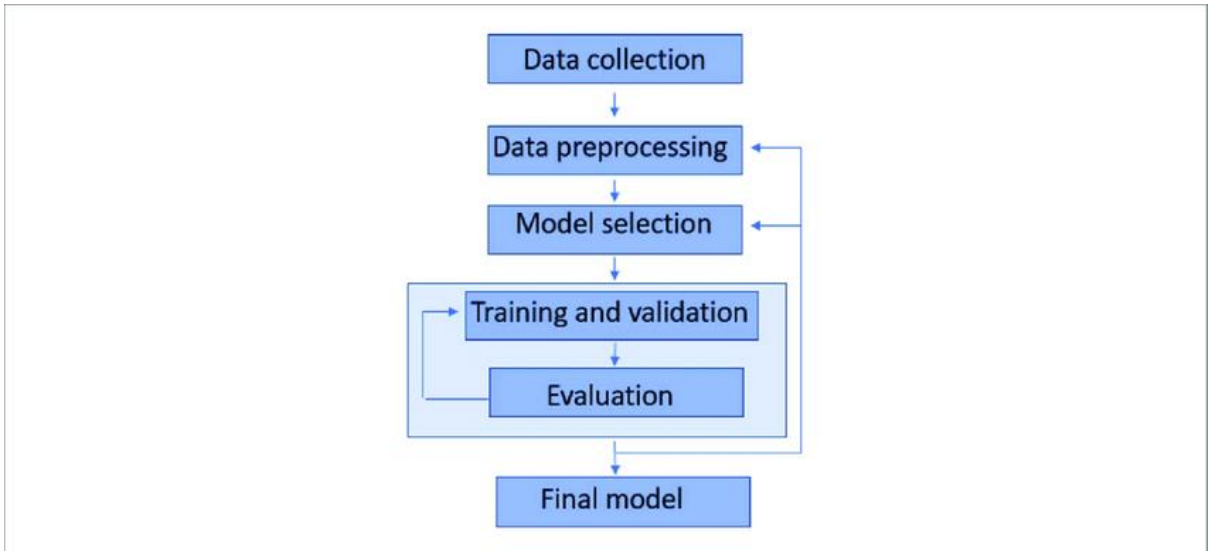
Section - P

- **<u>Problem Description:</u>**

The objective of this project is to design an end-to-end machine learning system that analyzes and predicts student academic performance using educational data. The system uses features such as study hours, attendance percentage, internal assessment scores, and previous academic results to understand factors influencing student performance. Data preprocessing techniques like data cleaning, normalization, and transformation ensure the dataset is accurate and consistent. **Exploratory Data Analysis (EDA)** is performed to study exam score distribution, detect outliers, and understand relationships between academic factors using visualizations such as histograms, box plots, and scatter plots. Correlation analysis and feature selection techniques like PCA, **forward selection**, and **backward selection** help identify the most important variables affecting performance. Machine learning models are developed to predict final exam scores using regression methods or classify students as pass or fail using classification algorithms.

In real life, such a system can help teachers and educational institutions identify students who are at risk of poor academic performance. Early prediction allows educators to provide targeted support, personalized learning strategies, and timely interventions to improve student outcomes and overall educational quality.

- **Steps of Implementation:**



1. Data Collection

In the first step, a suitable **student performance dataset** is collected. The dataset contains numerical attributes such as **study hours, attendance percentage, internal assessment marks, and previous academic scores**. This data serves as the input for the machine learning system.

2. Data Preprocessing

The collected data is cleaned and prepared before training the model. This includes **removing missing values, correcting inconsistent data, normalizing numerical values, and transforming features** which ensures that the dataset is reliable and improves the performance of machine learning models.

3. Model Selection

After preprocessing, appropriate **machine learning algorithms** are selected depending on the task. **Regression models** are used if the goal is to **predict exam scores**, while **classification models** are used to **classify students as pass or fail**.

4. Training and Validation

The dataset is split into **training and testing sets** in a ratio of **80:20**. The model is trained on the training data to learn patterns between features and student performance. Validation helps adjust model parameters and avoid overfitting.

5. Model Evaluation

The trained model is evaluated using appropriate performance metrics. For classification tasks metrics like **accuracy, precision, and recall** are used. For regression tasks metrics like **Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 Score** are used to measure how accurately the model predicts numerical values of student performance.

6. Final Model

The best-performing model is selected as the **final model**. This model can then be used by educators to **predict academic performance and identify students who may need additional support**.

- **<u>Expected Modules/ Libraries to be used:</u>**

1. **NumPy**
   - NumPy is used for **numerical computations and handling multi-dimensional arrays**.
   - It provides efficient mathematical operations required during data preprocessing and model calculations.

2. **Pandas**
   - Pandas is used for **data manipulation and analysis**.
   - It helps in reading datasets, cleaning data, handling missing values, and organizing data in structured formats like DataFrames.

3. **Matplotlib**
   - Matplotlib is a visualization library used to create **graphs and plots such as histograms, scatter plots, and line charts**.
   - It helps in understanding data patterns during Exploratory Data Analysis (EDA).

4. **Seaborn**
   - Seaborn is built on top of Matplotlib and is used for **advanced statistical visualizations** such as heatmaps, box plots, and correlation plots to analyze relationships between variables.
5. **Sk-learn**
   - Scikit-learn is a machine learning library used for **implementing algorithms such as regression and classification models**. It also provides tools for data splitting, feature selection, preprocessing, and model evaluation.
6. **Jupyter Notebook / Python**
   - Python programming language and Jupyter Notebook are used for **implementing the entire machine learning pipeline**, including data preprocessing, model training, and evaluation.

- **Expected Evaluation metrics and Plots to be used:**

1. **Accuracy**
   Accuracy shows how many predictions made by the model are correct compared to the total predictions. It helps determine how well the model can classify students as pass or fail.

2. **Precision**
   Precision tells us how many of the students predicted as passing are actually passing. It helps check the reliability of positive predictions made by the model.

3. **Recall**
   Recall measures how many of the actual passing students were correctly identified by the model. It helps ensure that students who are likely to pass or fail are not missed by the system.

Visualization Plots

1. **Histogram**
   a. A histogram will be used to see how student exam scores are distributed.
   b. It helps in understanding the overall pattern of scores in the dataset.

2. **Box Plot**
    a. A box plot helps in identifying outliers and understanding the spread of values such as exam scores or attendance percentages.
3. **Scatter Plot**
    a. A scatter plot will be used to observe relationships between variables.
    b. For example, how study hours or attendance may affect exam scores.
4. **Line graph**
    a. Line graphs are crucial in data analysis and machine learning for their ability to clearly display trends, patterns, and changes in data over a continuous progression, especially time.
    b. In model training, they are vital for visually tracking performance metrics to diagnose issues like overfitting or underfitting

- **<u>Required topics from the subject:</u>**

1. **Strings**
    - Strings are used to handle textual data in a program.
    - They help in storing and processing text-based information such as dataset labels, column names, or student details.

2. **Regex:**
    - Regular Expressions are patterns used to search, match, and manipulate text in strings.
    - They are commonly used for **data cleaning and validation**, such as removing unwanted characters or extracting useful information from text.

3. **Arrays:**

    - Arrays are used to store multiple values of the same type in a single variable. They help in managing and processing large amounts of data efficiently.

4.Function:

- **Functions are used to organize the code into smaller reusable blocks.**
- **They help in performing specific tasks such as data loading, preprocessing, model training, and prediction.**
- **Functions improve code readability, reduce repetition, and make debugging easier.**

- ## Platform Required:

The platforms that will be used for the implementation of this project are as follows:

- **Visual Studio Code**
- **Jupyter Notebook**
- **Kaggle (For Raw data)**
- **Gith & GitHub**

- ## Books and Link Sources:

- https://www.kaggle.com/datasets/nabeelqureshitiii/student-performance-dataset
- https://www.kaggle.com/datasets/spscientist/students-performance-in-exams/data
- https://www.kaggle.com/datasets/aljarah/xAPI-Edu-Data
- https://archive.ics.uci.edu/dataset/320/student+performance